

Temporary Tables

Store intermediate results in temporary storage.
during session

Database **drops all temp tables** once the session **ends**

When CTAS creates a table, it becomes part of permanent storage.

But temp tables, while acting similar to CTAS, creates run-time tables that only last as long as the session.

Session

Time between connecting and
disconnecting from the database.

w disconnecting from the database.

Syntax

SQL Server

SELECT

INTO # <table-name>

FROM

WHERE

to query from the temp table,

SELECT

FROM # <table-name>

WHERE

[Store it back again using CTAS if you like.
Hmm.]

Database Engine

Creates metadata → into catalog
Stores temp table → into temp section } all in
hard
disk

once session ends

clean up → both table in temp &
metadata

Use Cases

1. Intermediate results.

Extract intermediate result
In ETL pipelines → As temp table.
Transformation on
intermediate result.
Then load.

They can be dropped.

Automatic clean up.

not recommended

Automatic cleanup.

(Not recommended)

Need to check intermediate results
for debugging.

(ok for smaller projects I think.)

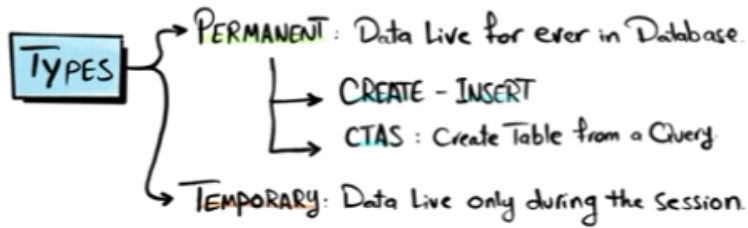
Rarely used by professionals

[Good for prototyping or other cases where auto-cleanup
is handy.

Summary

TABLES

Structured Collection of Data like spreadsheet (Columns & Rows)



CTAS USE CASES

- Optimize Performance: Persist Complex SQL Logic in Table
- Creating Snapshot: to analyse Bugs and data issues.

Advantage of TEMP Tables Automatic cleanup of data after session ends

Summary of all 5 techniques

Baraa's utility Ranking

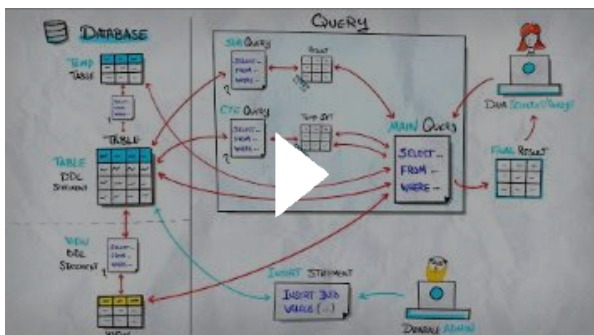
- Subquery _____ 3
- CTE _____ 2
- VIEW _____ 1
- CTAS _____ 4
- TEMP TABLES. _____ 5



	SUBQUERY	CTE	TMP	CTAS	VIEW
STORAGE	MEMORY 		DISK 		X NO STORAGE
LIFE TIME	TEMPORARY			PERMANENT	
WHEN DELETED	END OF QUERY		END OF SESSION	DDL - DROP	
SCOPE	SINGLE - QUERY		MULTI - QUERIES		
REUSABILITY	LIMITED 1 PLACE - 1 QUERY	LIMITED Multi PLACES - 1 QUERY	MEDIUM Multi QUERIES During Session	HIGH MULTI QUERIES	
UP2DATE					 

Big Picture

[SQL Subquery vs CTE vs Views vs CTAS vs TEMP | #SQL Course 32](#)



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